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Direct Numerical Simulation of Flow over Periodic Hills up to $\mathrm{Re}_H = 10{,}595$

Benjamin Krank[1] · **Martin Kronbichler**[1] · **Wolfgang A. Wall**[1]

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Abstract We present fully resolved computations of flow over periodic hills at the hill-Reynolds numbers $\mathrm{Re}_H = 5{,}600$ and $\mathrm{Re}_H = 10{,}595$ with the highest fidelity to date. The calculations are performed using spectral *incompressible* discontinuous Galerkin schemes of 8$^{\text{th}}$ and 7$^{\text{th}}$ order spatial accuracy, 3$^{\text{rd}}$ order temporal accuracy, as well as 34 and 180 million grid points, respectively. We show that the remaining discretization error is small by comparing the results to h- and p-coarsened simulations. We quantify the statistical averaging error of the reattachment length, as this quantity is widely used as an 'error norm' in comparing numerical schemes. The results exhibit good agreement with the experimental and numerical reference data, but the reattachment length at $\mathrm{Re}_H = 10{,}595$ is predicted slightly shorter than in the most widely used LES references. In the second part of this paper, we show the broad range of capabilities of the numerical method by assessing the scheme for underresolved simulations (implicit large-eddy simulation) of the higher Reynolds number in a detailed h/p convergence study.

Keywords Periodic hill flow · Incompressible Navier–Stokes equations · Direct numerical simulation · Large-eddy simulation · High-order discontinuous Galerkin

✉ Benjamin Krank
krank@lnm.mw.tum.de

Martin Kronbichler
kronbichler@lnm.mw.tum.de

Wolfgang A. Wall
wall@lnm.mw.tum.de

[1] Institute for Computational Mechanics, Technical University of Munich, Boltzmannstr. 15, 85748 Garching, Germany

1 Introduction

Flow over periodic hills according to, e.g., Temmerman et al. [1] and Fröhlich et al. [2], has in recent years become one of the most widely used test cases for the validation and assessment of computational fluid dynamics codes and turbulence models. Its popularity lies in its simplicity regarding simulation setup and boundary conditions on the one hand and complexity with respect to flow phenomena and turbulence modeling on the other hand. The flow configuration consists of streamwise periodically arranged smoothly curved hills at the lower wall, a plane boundary at the upper wall, and periodic boundary conditions in spanwise direction. A major advantage of this setup is the periodicity in streamwise direction, which does not require the definition of synthetic inflow conditions, and therefore simplifies the reproducibility of the results. The developed turbulent flow is depicted in Fig. 1 and shows several challenging and highly relevant flow phenomena, which makes this example particularly interesting as a benchmark case, such as: separation from a curved surface, recirculation in the wake of the hill, a sharp shear layer, strong pressure gradients, and flow reattachment. The flow was used as a test case, e.g., for wall-resolved large-eddy simulation (LES) [3–7] and wall models [3, 8–11]. In the context of Reynolds-averaged Navier–Stokes (RANS) and hybrid RANS/LES, this flow was analyzed rigorously within the European initiative "Advanced Turbulence Simulation for Aerodynamic Application Challenges" (ATAAC), of which the final report by Jakirlić is available at [12].

1.1 Previous reference data

Several reference databases exist for this flow and an overview is given in Tables 1 and 2 for two Reynolds numbers. The flow has been investigated numerically via LES by Fröhlich et al. [2] at a Reynolds number $\mathrm{Re}_H = Hu_b/\nu = 10{,}595$, based on the hill height H,

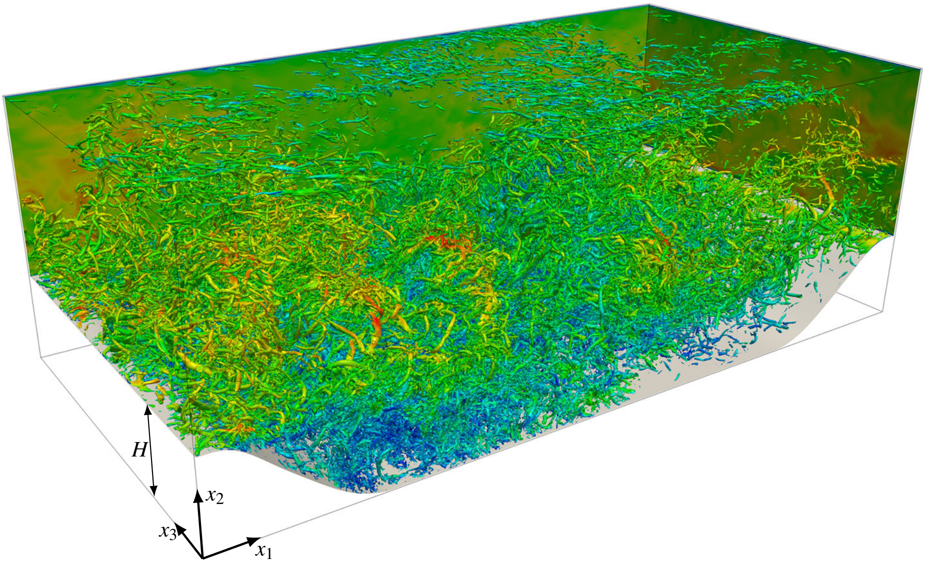

Fig. 1 Visualization of the turbulent flow structures at $\mathrm{Re}_H = 10{,}595$ via iso-surfaces of the Q-criterion colored by velocity magnitude. Red indicates high and blue low velocity

Table 1 $\mathrm{Re}_H = 5{,}600$: DNS and under-resolved DNS (URDNS) simulation cases in comparison to reference data, including polynomial degree k, order of accuracy $(k+1)$, total number of grid points N, sampling time Δt_{aver} in number of flow-through times T^*, separation and reattachment length $x_{1,\mathrm{sep}}$ and $x_{1,\mathrm{reatt}}$

Label	Grid points	k	Order	N	Type	$\frac{\Delta t_{\mathrm{aver}}}{T^*}$	$\frac{x_{1,\mathrm{sep}}}{H}$	$\frac{x_{1,\mathrm{reatt}}}{H}$
URDNS 5600	448×224×224	6	7	22.5M	URDNS	61	0.16	4.82 ± 0.09
DNS 5600	512×256×256	7	8	33.6M	DNS	61	0.17	5.04 ± 0.09
BPRM LES [13]	280×220×200	–	2	13.1M	LES	140	0.18	5.09
BPRM DNS [13]	765×750×404	–	2	231M	DNS	38	0.18	5.14
RM Exp [14]	–	–	–	–	Exp.	–	–	4.83

The number of nodes of the present DG solver is obtained by the number of cells in each spatial direction times the number of nodes per cell $(k+1)$ in each direction. All cases are incompressible

bulk velocity above the hill u_b, and the kinematic viscosity ν. The width of the domain in spanwise direction was studied and a value of $4.5H$ was found sufficient, which has since been used in most studies. The simulation was carried out twice using the same grid, but with two independent second order accurate finite volume codes and these cases are in Table 2 labeled as FMRTL R1 and R2, respectively, corresponding to RUN1 and RUN2 in [2]. The reference data base has been complemented by Breuer et al. [13], who performed simulations over a wide range of Reynolds numbers, starting from $\mathrm{Re}_H = 700$ and up to $\mathrm{Re}_H = 10{,}595$. Again, second order accurate finite volume schemes were used and the simulations are fully resolved (direct numerical simulation, DNS) up to $\mathrm{Re}_H = 5{,}600$ and well-resolved (LES) at $\mathrm{Re}_H = 10{,}595$. These cases are referred to as BPRM (DNS/LES) in Tables 1 and 2. Finally, Rapp and Manhart [14] conducted water-channel experiments of the Reynolds numbers $\mathrm{Re}_H = 5{,}600$, $\mathrm{Re}_H = 10{,}600$, $\mathrm{Re}_H = 19{,}000$, and $\mathrm{Re}_H = 37{,}000$, labeled as RM Exp in Tables 1 and 2. The data of Breuer et al. [13] and Rapp and Manhart [14] is available in the ERCOFTAC QNET-CFD Wiki contributed by Rapp et al. [12] alongside a detailed test case description. Further experimental data was recently acquired at the Reynolds numbers $\mathrm{Re}_H = 8{,}000$ and $\mathrm{Re}_H = 33{,}000$ [15].

Table 2 $\mathrm{Re}_H = 10{,}595$: DNS and under-resolved DNS (URDNS) simulation cases in comparison to reference data, including polynomial degree k, order of accuracy $(k+1)$, total number of grid points N, Mach number Ma/incompressible (inc.), sampling time Δt_{aver} in number of flow-through times T^*, separation and reattachment length $x_{1,\mathrm{sep}}$ and $x_{1,\mathrm{reatt}}$

Label	Grid points	k	Order	N	type	Ma	$\frac{\Delta t_{\mathrm{aver}}}{T^*}$	$\frac{x_{1,\mathrm{sep}}}{H}$	$\frac{x_{1,\mathrm{reatt}}}{H}$
URDNS 10595	768×384×384	5	6	113M	URDNS	inc.	61	0.19	4.57 ± 0.06
DNS 10595	896×448×448	6	7	180M	DNS	inc.	61	0.20	4.51 ± 0.06
FMRTL R1 [2]	196×128×186	–	2	4.7M	LES	inc.	55	0.20	4.56
FMRTL R2 [2]	196×128×186	–	2	4.7M	LES	inc.	55	0.22	4.72
BPRM LES [13]	280×220×200	–	2	13.1M	LES	inc.	140	0.19	4.69
DM [5]	384×192×192	7	8	14M	ILES	0.1	25	0.20	4.37
BPP [3]	800×500×500	–	7	200M	DNS	0.2	15[a]	–	4.5
RM Exp [14]	–	–	–	–	Exp.	inc.	–	–	4.21

The number of nodes of the present DG solver is obtained by the number of cells in each spatial direction times the number of nodes per cell $(k+1)$ in each direction

[a] Private communication with Ponnampalam Balakumar

The available reference data shows significant differences, in particular between the experimental and the numerical DNS/LES data; for example, the reattachment lengths range from $4.83H$ (experiments) up to $5.14H$ (DNS) in the case $\mathrm{Re}_H = 5{,}600$ and from $4.21H$ (experiments) to $4.72H$ (LES) in the case $\mathrm{Re}_H = 10{,}595$.

Further simulations with higher resolution and high-order accurate schemes have been presented in [3] (labeled BPP in Table 2) and [5] (labeled DM in Table 2), however, these studies employed *compressible* solvers at low Mach numbers and comparably short averaging times of 15 flow-through times[1] and 25 flow-through times, respectively, which renders them unsuitable as reference data [5].

1.2 Scope and outline of the present study

The primary goal of the present article is to shed light on the differences visible in the previous reference data sets, to discuss sources of error, and to provide a new set of reference data for the Reynolds numbers $\mathrm{Re}_H = 5{,}600$ and $\mathrm{Re}_H = 10{,}595$. Possible weaknesses of the previous numerical reference data may be:

- All reference computations were carried out using similar numerical schemes of second order spatial accuracy, which consequently also have similar dissipation and dispersion properties. In order to provide new insights into the flow cases, we perform simulations using an entirely different scheme of high-order spatial accuracy with very low dissipation and dispersion characteristics.
- The averaging times of the numerical reference data lie in a broad range of approximately $\Delta t_{\mathrm{aver}}/T^* = 38$ to $\Delta t_{\mathrm{aver}}/T^* = 140$ flow-through times, based on the sampling time Δt_{aver} and a single flow-through time $T^* = 9H/u_b$. In this work, we investigate the issue of the sampling time by estimating the statistical averaging error and show that the latter is nonnegligible for common averaging times.
- The available skin friction and pressure coefficient curves show numerical oscillations due to the spatial discretization. The simulations conducted within this study provide these quantities with a higher quality.

In addition to the high resolution, we further investigate the quality of the DNS results by providing a detailed h/p convergence study towards the DNS at the higher Reynolds number. Since these coarser simulations do not resolve all turbulent scales, they may be classified as implicit large-eddy simulation (ILES). To this end, despite the low dissipation of the present numerical scheme when all scales are resolved, the numerical fluxes provide sufficient subgrid dissipation for such underresolved simulations. Besides serving as a convergence analysis for the DNS, the results also present new data with respect to the sensitivity of the method regarding a variation of the grid size and polynomial degree of the cells in underresolved simulations.

The remainder of this article is organized as follows. In Section 2, we outline the numerical method used for the current computations and discuss its resolution power. In Section 3, we present the simulation setup and the meshes, and discuss the resolution of the DNS. The DNS results are shown in Section 4. The ILES h/p convergence study is presented in Section 5 and conclusions close the article in Section 6.

[1] Private communication with Ponnampalam Balakumar.

2 Spectral *Incompressible* Discontinuous Galerkin Solver

We give a brief overview of the numerical scheme considered and its basic characteristics.

2.1 Numerical scheme

The present DNS are performed using the spectral *incompressible* discontinuous Galerkin scheme INDEXA [16]. In the present numerical method, the time integration of the *incompressible* Navier–Stokes equations is based on the high-order semi-explicit dual-splitting method by Karniadakis et al. [17], splitting each time step into three substeps. The nonlinear convective term is first treated explicitly in order to avoid nonlinear iterations. The pressure is determined by solving a Poisson equation, and the result is used to make the velocity divergence-free in a local projection. In the third step, the viscous effects are taken into account in an implicit Helmholtz-like equation. Throughout this work, we choose the temporal discretization parameters such that third order accuracy in time is achieved both in velocity and pressure [16]. The maximum allowable time step increments are computed adaptively in each time step using a Courant number of $\mathrm{Cr} = 0.09$ and the CFL condition presented in [10]. Note that this CFL condition includes an inverse scaling of the time step size by a factor of $k^{1.5}$. Nodal tensor-product elements based on the Gauss–Lobatto points of polynomial degree k are used for the spatial discretization of the velocity and pressure, which have $(k+1)^3$ grid points per cell, i.e., $k+1$ per spatial direction in each cell. The cells are connected using appropriate fluxes. For the convective term, we choose the local Lax–Friedrichs flux, and the symmetric interior penalty method for the second derivatives [18]. To ensure mass conservation in the context of the discontinuous Galerkin discretization and stability for small time increments, we apply the variant *V3c* in [16]. Therein, the right-hand sides of the pressure Poisson and projection steps are considered in the partially integrated form and a consistent supplementary term reduces the local continuity error; see also [19]. A supplementary stability analysis of the resulting formulation via energy arguments has been performed in [20] and confirms the results from [16]. All integrals in the weak forms of the *incompressible* scheme are computed exactly on affine cells according to [16] (denoted ‘overintegration’, see, e.g., [21, 22]), obviating explicit dealiasing or filtering. To this end, the convective term is evaluated using Gaussian quadrature on $\left\lfloor \frac{3k}{2} \right\rfloor + 1$ points per coordinate direction and all remaining terms on $k+1$ points per coordinate direction. The scheme exhibits a spatial order of accuracy of $k+1$ both in velocity and pressure [16]. Although the spatial order of accuracy used for the present simulations is much higher than the temporal order of accuracy, the discretization error is dominated by the spatial discretization. This conclusion was verified from a comparison of results obtained with temporal schemes of second and third order. An explanation for this behavior is that CFL stability restrictions force small enough time steps.

The solver is implemented in a fully matrix-free manner, including the pressure Poisson multigrid algorithm [23, 24], using the fast kernels by Kronbichler and Kormann [25] within the deal.II finite element library [26]. The computational time per time step and degree of freedom of this scheme is almost independent of the polynomial degree up to approximately $k = 6$. In addition, the algorithms are highly scalable; the present code has been scaled up to 147,456 CPU cores and 34.4 billion degrees of freedom on SuperMUC of the Bavarian supercomputing center LRZ in Garching, Germany [16]. These characteristics make INDEXA ideally suited for DNS of turbulent flows. The solver has been verified and validated using laminar and turbulent flow examples, including DNS of turbulent channel flow [16]. First results of ILES for the periodic hill flow at a Reynolds number of $\mathrm{Re}_H = 10{,}595$ were presented in [6] using this scheme.

2.2 Resolution power of spectral DG methods

The basic characteristics of high-order discontinuous Galerkin schemes have recently been investigated via dispersion-diffusion analysis, for example by Gassner and Kopriva [27] and Moura et al. [28, 29], with regard to their applicability to turbulent flows. The analyses show quantitatively how higher order DG schemes exhibit much lower dispersion and dissipation characteristics than lower order methods and maintain these over a broader wave number range. Therefore, high-order DG schemes have been widely used for DNS, for example in [7, 16, 30–33].

Based on the dispersion-diffusion analysis, [28, 29] introduces a measure for the efficiency of a numerical scheme, the 1% rule, which specifies the minimum number of grid points per wave length in which the amplitude of a scale is dissipated by less than 1% while being convected by a distance of $h_e/(k+1)$. As an example, a second-order DG scheme ($k = 1$) would require 8.2 nodes per wave length (in 3D [28]), whereas 3.4 nodes would suffice for a scheme of 6$^{\text{th}}$ order accuracy, 3.2 nodes for a scheme of 7$^{\text{th}}$ order accuracy, and 3.1 nodes for a scheme of 8$^{\text{th}}$ order accuracy. This analysis allows the conclusion that a DG scheme introduces almost zero numerical dissipation if all scales are larger than the limit defined by the 1% rule, and that high polynomial orders are much more efficient in order to achieve this goal. In practice, we argue that schemes of $k = 5$ to $k = 7$ are very efficient for DNS, since, for larger polynomial degrees, the resolution power saturates, the solution of linear systems becomes more expensive per degree of freedom [24], and the inverse scaling of the maximum time step size with $k^{1.5}$ considering the present scheme requires a relatively large number of time steps [34].

Another characteristic of higher-order DG methods is the stronger damping of scales that are just marginally larger than the grid filter size with increasing degree [27, 29]. This numerical dissipation, which is necessary to stabilize the scheme in underresolved simulations, may in fact be appropriate to model the unresolved turbulent scales, according to the concept of ILES. Moderate and high-order DG schemes have been extensively used for ILES, see for example [4, 16] and the references therein. A literature review of ILES studies and a discussion on whether additional explicit subgrid terms should be used is given in Section 5 in the context of underresolved simulations.

3 Simulation Setup, Resolution Requirements, and Mesh

We give an overview of the simulation setup, boundary conditions, and the mesh used for the DNS simulations.

3.1 Simulation setup

We consider a domain of the dimensions $9H \times 3.036H \times 4.5H$ in streamwise, wall normal, and spanwise direction, respectively. Periodic boundary conditions are applied in streamwise and spanwise direction and no-slip condition at the lower and upper wall. The flow is driven as in [2] by a pressure gradient, represented by a body force $\boldsymbol{f} = (f_1, 0, 0)^T$, where

f_1 is constant in space. The volume flux $Q = \int_\Omega u_1 \, d\Omega/9H$, and thus the Reynolds number, is kept constant in time by a control algorithm, which is similar to the one proposed in Benocci and Pinelli [35], and detailed in Appendix A. The simulations are run for about 84 flow-through times T^* of which statistics are sampled during $61T^*$ after the initial transient. The statistics are also averaged over the homogeneous spanwise direction and more than 100,000 samples are used in the averaging process for the highly resolved simulation cases in order to minimize the error stemming from the statistical postprocessing.

3.2 Meshes

For the DNS cases, we use a spatial discretization with 64×32×32 cells with a polynomial degree of $k = 7$, yielding 512×256×256 grid points and a spatial accuracy of 8$^{\text{th}}$ order, for the case $\text{Re}_H = 5{,}600$, and a grid of 128×64×64 cells with a polynomial degree of $k = 6$, yielding 896×448×448 grid points and a spatial accuracy of 7$^{\text{th}}$ order, for the case $\text{Re}_H = 10{,}595$. These meshes are displayed in Fig. 2. The grid is equally spaced in streamwise and spanwise direction and mildly stretched towards the walls in order to improve the near-wall resolution. In order to guarantee a highly accurate representation of the boundary, the curved hill geometry is represented by the same polynomial degree as the solution with mapping facilities provided by the deal.II library [26]. The vertical grid lines are aligned parallel to the vertical axis and it is noted that orthogonality of the grid is not relevant in the context of DG as the stencils used herein are truly three-dimensional. A second grid is considered for each Reynolds number with a very similar layout, but with one degree lower (p-coarsening). The DNS cases are in the following labeled DNS 5600 and DNS 10595, respectively, and the p-coarsened cases URDNS 5600 and URDNS 10595, URDNS denoting under-resolved DNS, according to Tables 1 and 2. In Section 5, 12 supplementary h/p-coarsened meshes

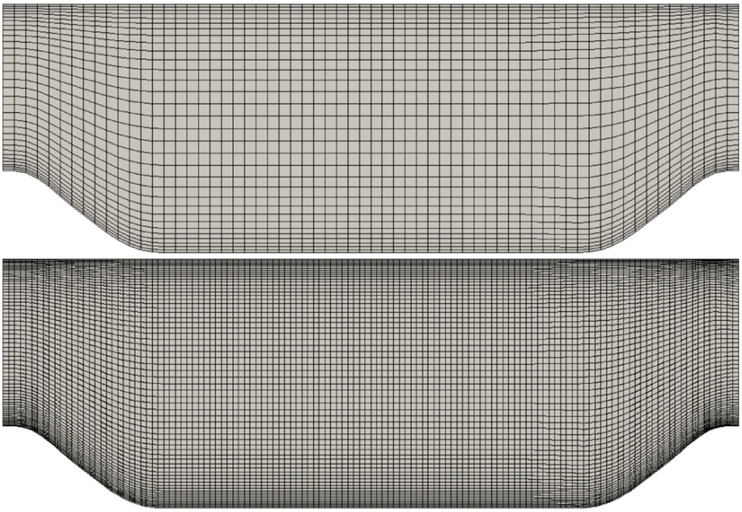

Fig. 2 Grid for DNS. Top: DNS 5600. Bottom: DNS 10595. In each grid cell, the solution is represented by polynomials of 7$^{\text{th}}$ ($\text{Re}_H = 5{,}600$) and 6$^{\text{th}}$ ($\text{Re}_H = 10{,}595$) degree and the scheme exhibits an order of accuracy of polynomial degree plus one

are considered for the case $\mathrm{Re}_H = 10{,}595$. A discussion of the computational cost of the DNS cases is included in Appendix B.

3.3 Assessment of resolution

The mesh resolution is evaluated using criteria for the near-wall region and the bulk flow as in [2, 13]. Figure 3 shows the location of the first off-wall Gauss–Lobatto point, corresponding to the first off-wall grid point at the wall inside the cell, Δx_2^+, as well as the dimensionless streamwise and spanwise grid-spacings $\Delta x_1^+ = \Delta x_3^+$. These wall coordinates are defined as $\Delta x_2^+ = \Delta x_2 u_\tau/\nu$ with the friction velocity $u_\tau = \sqrt{\tau_w/\rho}$ and the density ρ, and $\Delta x_1^+ = \Delta x_3^+ = \Delta x_e u_\tau/\nu(k+1)$. The quantity Δx_2^+ is below 0.70 at the upper wall for both DNS cases. At the lower wall, Δx_2^+ exhibits a characteristic peak on the windward side of the hill crest with a maximum of $\max(\Delta x_2^+) = 0.86$. In the streamwise and spanwise directions, the distribution is similar with maximum values of $\max(\Delta x_1^+) = 7.2$ at the upper wall and a more pronounced peak at the lower wall of up to $\max(\Delta x_1^+) = 13.7$. This resolution is well sufficient to fully resolve the velocity gradient and the turbulent scales at the wall.

The resolution of the bulk flow is assessed by comparing the grid filter width $h = h_e/(k+1)$ with the resolved Kolmogorov length $\eta = (\nu^3/\tilde{\varepsilon})^{1/4}$. The grid size is defined based on a cell volume V as $h_e = V^{1/3}$ neglecting direction-dependence, however, the cells in the bulk flow are almost cubic, since $\Delta x_1 = \Delta x_3$ throughout and Δx_2 is in a similar range. The dissipation rate is computed via the resolved velocity field vector $\boldsymbol{u}$ as $\tilde{\varepsilon} = \nu((\nabla \boldsymbol{u})_{ij}(\nabla \boldsymbol{u})_{ij})$ in index notation (see [36] for further details) and averaged over time and the spanwise direction. In Fig. 4 (top), we show averaged profiles of η at nine streamwise stations, located at $x_1/H = 0.05, 1, 2, 3, 4, 5, 6, 7, 8$, and compare the respective DNS and URDNS cases. As only small differences are visible despite the varying resolution, it is assumed that the subgrid scale contribution to the dissipation rate is negligible. In Fig. 4 (bottom), the ratio h/η is plotted over all nine stations and iso-lines at values of 0, 2, and 4 are included as a reference. The value of h/η is located between 2 and 4 with peaks in

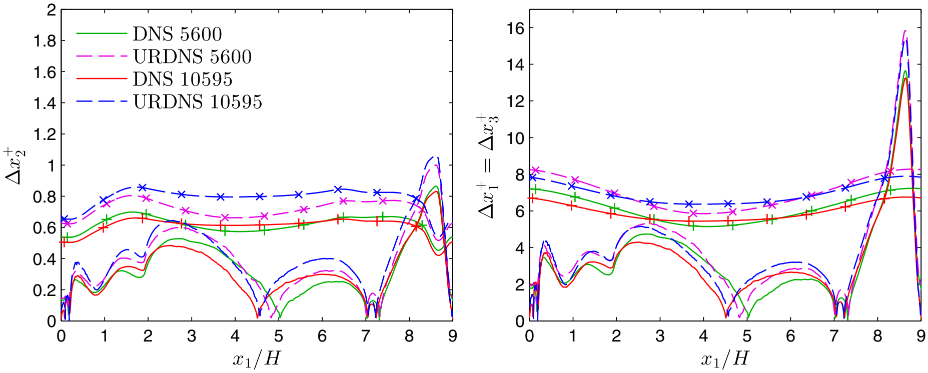

Fig. 3 Location of first off-wall Gauss–Lobatto point Δx_2^+ (left) as well as normalized grid spacings in streamwise and spanwise direction $\Delta x_1^+ = \Delta x_3^+$ (right). The curves corresponding to the upper wall include markers

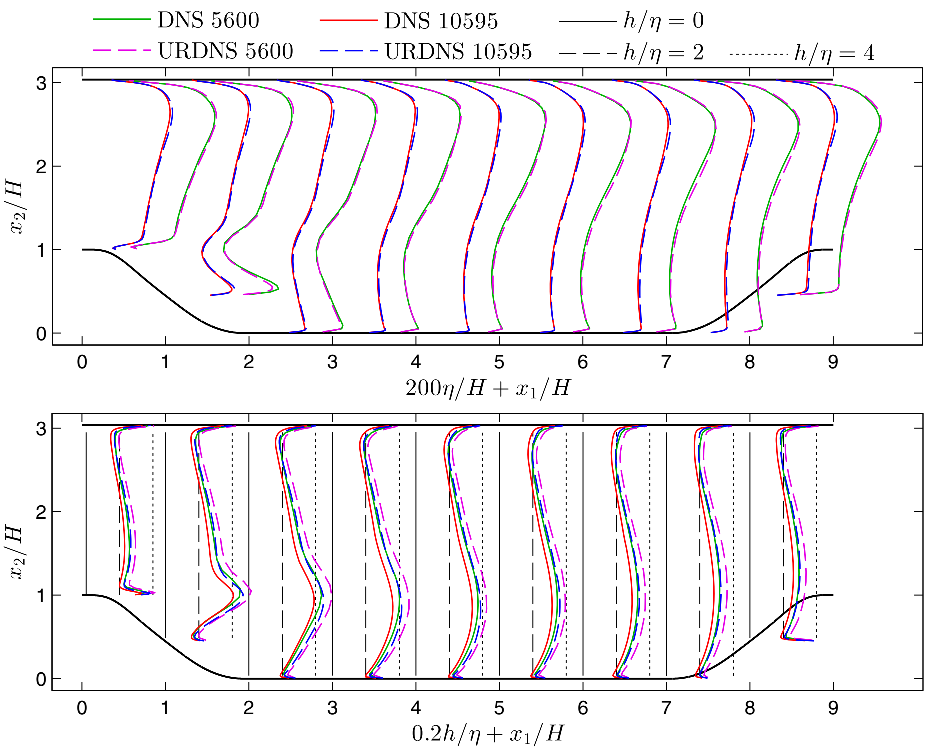

Fig. 4 Assessment of resolution: Kolmogorov length through resolved dissipation (top) and ratio of grid size to Kolmogorov length (bottom)

the shear layer of 4.48 in the case DNS 5600 and 4.07 in the case DNS 10595. Pope [36] specifies the 24η-range as the length scale where most dissipation occurs and Moin and Mahesh [37] state that most dissipation is resolved above 15η, and the latter publication lists DNS examples with a resolution around $h/\eta \sim 4$. Considering the spectral characteristics of the present scheme, we regard the resolution of the DNS cases sufficient to be denoted *direct numerical simulation*. The URDNS cases exhibit $\max(h/\eta) = 5.15$ and also use a scheme with one degree lower, which exhibits larger dissipation and dispersion errors, and are therefore denoted *under-resolved DNS*.

4 Results of Direct Numerical Simulation

We present the results for the skin friction and pressure coefficients as well as velocity statistics separately for each Reynolds number and compare the DNS with the available reference data. While it is the aim of this study to provide new reference data for both Reynolds numbers, the results of the present work are first compared to highly resolved reference data at the lower Reynolds number to validate the simulation setup. Subsequently, some more controversial issues are discussed regarding the higher Reynolds number. The last part of this section presents an analysis of the statistical sampling error including the confidence interval of the reattachment length.

4.1 $\mathbf{Re}_H = 5{,}600$

4.1.1 $Re_H = 5{,}600$: *skin friction and pressure coefficients*

We commence the discussion of the simulation results by considering the skin friction and pressure coefficients

$$c_f = \frac{\tau_w}{\frac{1}{2}\rho u_b^2}, \qquad c_p = \frac{p - p_{\mathrm{ref}}}{\frac{1}{2}\rho u_b^2},$$

with the wall shear stress τ_w, the pressure p and a reference pressure p_{ref} at the probe location at $x_1 = 0$ on the bottom wall. Details on how the wall-parallel component of the stress is computed on the hill shape at the lower boundary are given in Appendix C. The results for both coefficients at the upper and lower wall are shown in Fig. 5 and the curves are compared to the DNS and LES results of [13], labeled BPRM DNS and BPRM LES, respectively, according to Table 1. Regarding the skin friction, the case URDNS 5600 overpredicts the skin friction to a minor extent between $x_1/H = 4$ and $x_1/H = 7$ in comparison to the cases DNS 5600 and BPRM DNS. The two DNS cases, DNS 5600 and BPRM DNS, are in very good agreement overall. However, the case BPRM DNS exhibits oscillations near the peak of the skin friction at the hill crest with an amplitude of up to approximately 10% of the magnitude, whereas the case DNS 5600 shows smooth data. The simulation BPRM LES predicts the skin friction in the region $x_1/H = 4$ to $x_1/H = 7$ at the bottom wall slightly lower than the two DNS, but the agreement is on a very high level overall. The profiles of the skin friction at the lower wall predict the presence of two recirculation bubbles in accordance with the reference simulations [13]. The primary recirculation zone separates around $x_{1,\mathrm{sep}}/H = 0.17$ and the reattachment point lies near $x_{1,\mathrm{reatt}}/H = 5.0$, determined via the zero-crossings of the skin friction. These locations are in good agreement with the reference data and the exact values are compared in Table 1. Furthermore, a separate section of this article is devoted to a discussion of the quality of the predicted reattachment point, see Section 4.3. The second recirculation bubble is located just in front of the hill at the windward side, between $x_1/H = 7.04$ and $x_1/H = 7.31$ considering the results of the present DNS.

Regarding the pressure coefficient, there are more significant differences between the two DNS cases, DNS 5600 and BPRM DNS, and the underresolved cases, URDNS 5600 and BPRM LES. The largest deviations between the simulation cases are visible in the range $x_1/H = 0$ to $x_1/H = 1$ at the lower wall, where the pressure sees a strong positive pressure gradient. The length of this zone with positive pressure gradient varies significantly for the different flow cases. Since the pressure at $x_1/H = 0$ at the lower wall is used as the reference pressure, this discrepancy shifts the pressure curves considerably. We argue therefore that the pressure at $x_1/H = 0$ at the upper wall should be used as the reference pressure in future work, which has shown to be much more robust with respect to the resolution near the hill crest. This pressure value is unfortunately not available for all reference data sets considered herein, but the pressure coefficient curves at the upper wall of the present data set are included in the downloadable data files in [38]. However, the two DNS cases, DNS

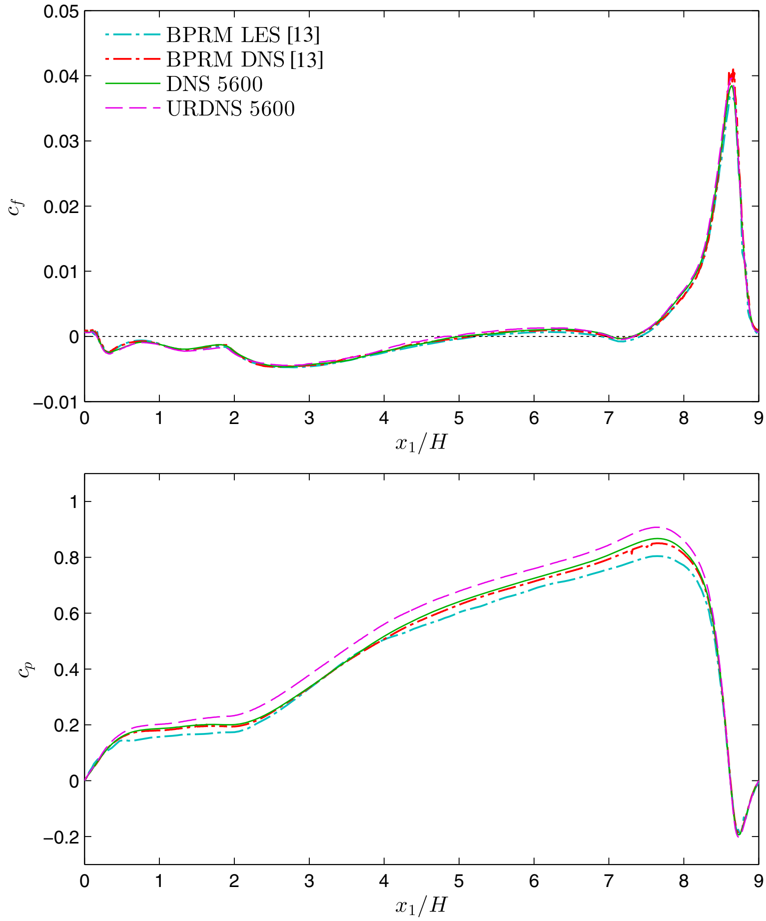

Fig. 5 $\mathrm{Re}_H = 5{,}600$: Skin friction (top) and pressure (bottom) coefficients at the lower wall

5600 and BPRM DNS, show a very high level of agreement, confirming the high fidelity of these cases.

4.1.2 $Re_H = 5{,}600$: velocity statistics

The same cases are further compared via profiles of the streamwise and vertical mean velocity, Reynolds shear stress and turbulence kinetic energy in Fig. 6. The data is plotted at nine streamwise stations, which are located at $x_1/H = 0.05$, 1, 2, 3, 4, 5, 6, 7, and 8. Additionally, the simulation results are compared to the experimental data available [14], labeled RM Exp, including the mean velocity profiles and the Reynolds shear stress. The results of the mean velocity profiles of all four simulation cases are in excellent agreement, and the accordance of the streamwise velocity with the experimental data, RM Exp, is good. Regarding the vertical velocity, the experimental data exhibits the largest deviations from

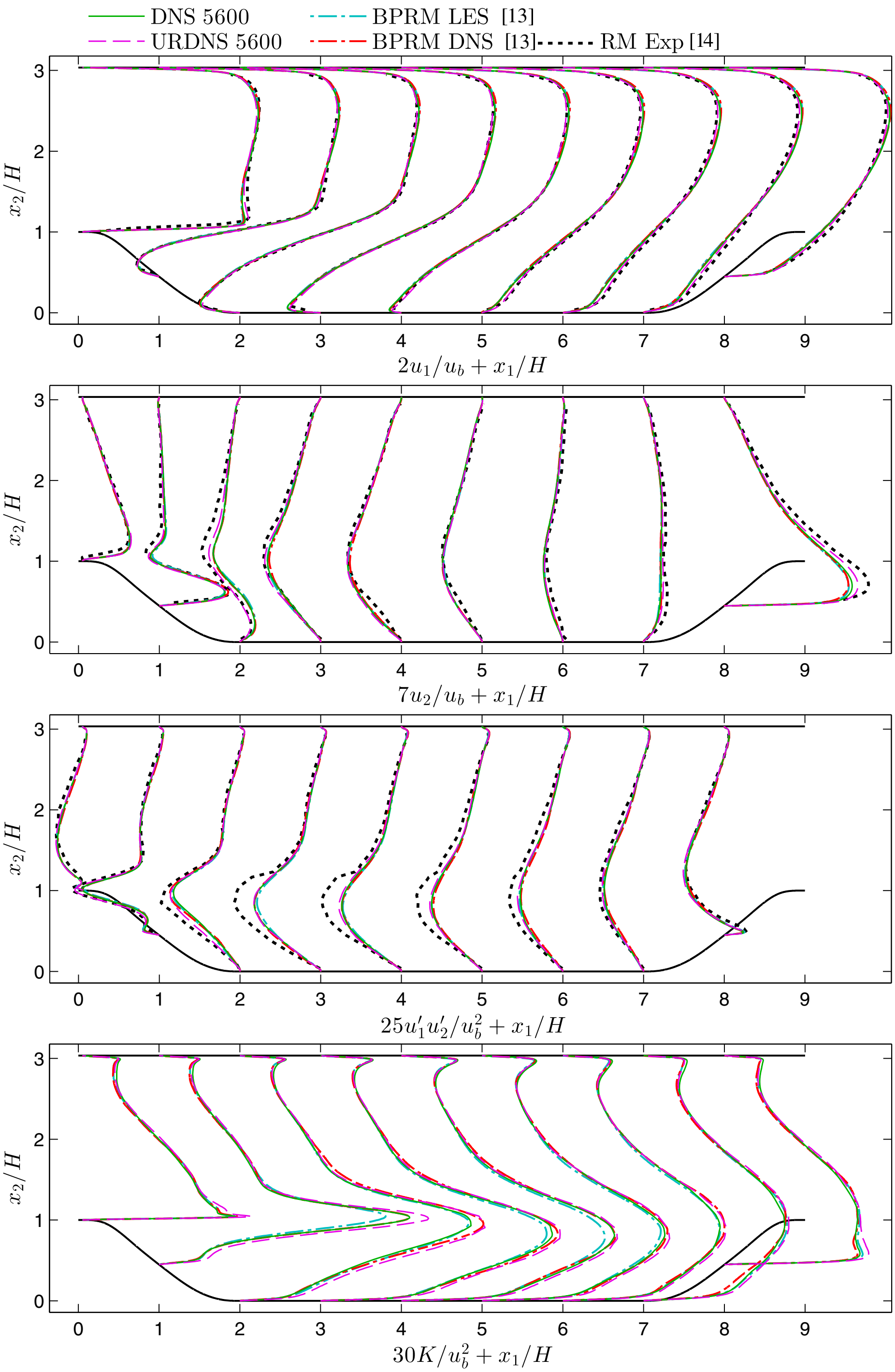

Fig. 6 $\mathrm{Re}_H = 5{,}600$: Profiles of mean velocity in streamwise and vertical direction u_1 and u_2, Reynolds shear stress $u_1'u_2'$, as well as turbulence kinetic energy $K = 1/2(u_1'u_1' + u_2'u_2' + u_3'u_3')$ (from top to bottom)

the other data sets. At the fourth ($x_1/H = 2$) and last station ($x_1/H = 8$), the case URDNS 5600 also shows differences to the three other simulation cases.

With respect to the Reynolds shear stress, almost no difference may be observed between the four numerical data sets. The experimental data shows values of much higher magnitude than the simulation cases. This effect may be due to the different spanwise extension of the experimental and numerical domains [15].

Finally, the profiles of the turbulence kinetic energy show pronounced differences in the shear layer region above the recirculation zone. The two DNS, DNS 5600 and BPRM DNS, exhibit excellent accordance up to $x_1/H = 1$. The case BPRM LES shows an underprediction of the profile in the shear layer in comparison to the two DNS. This behavior may indicate a minor underresolution in that region. The case URDNS 5600 may also be slightly under-resolved in that region, as the peak in the shear layer is higher than the DNS. At $x_1/H = 2$, the cases BPRM DNS and DNS 5600 deviate significantly. One reason for this behavior may be the significant change in resolution in streamwise direction of the case BPRM DNS in this area, which may come along with artificial numerical dissipation, as the propagation of small scales through streamwise stretched grids has a dissipating effect, which has for example been investigated for standard second order central differences in [39]. In addition, the mesh of the case BPRM DNS shows high aspect ratios in the shear layer of $\Delta x_1/\Delta x_2 > 10$ (see [13]), while the resolution was assessed by comparing the Kolmogorov length to the cube root of the cell volume, neglecting direction dependence. As a result, the streamwise cell sizes may be too coarse to represent a DNS in the shear layer at $x_1/H = 2$ and beyond. The differences of the numerical data are less significant at the downstream stations.

From this discussion, we conclude that

- the two DNS cases, DNS 5600 and BPRM DNS, are in very good agreement overall,
- the new DNS data does not show oscillations in the skin friction,
- the new DNS data has a longer sampling time than the case BPRM DNS, which reduces the uncertainty of the quantities (see Section 4.3).

4.2 $\mathrm{Re}_H = 10{,}595$

4.2.1 $Re_H = 10{,}595$: *skin friction and pressure coefficients*

In Fig. 7, we compare the profiles of the skin friction and pressure coefficients of the present simulations DNS 10595 and URDNS 10595 to the cases BPRM LES [13], FMRTL R2 [2], and DM [5], according to Table 2. Regarding the skin friction, we notice significant disagreement between $x_1/H = 7$ and $x_1/H = 9$: in the acceleration zone at the windward side of the hill, the simulation cases DNS 10595 and URDNS 10595 show very good agreement with BPRM LES in that region, whereas the cases FMRTL R2 and DM predict a significantly smaller skin friction, but agree well with each other. This difference may be due to a different postprocessing of the wall shear stress. We have found that our DNS results agree well with the cases FMRTL R2 and DM in that region if only the u_1-component of the wall shear stress, $\tau_{w,1}$, is taken into account according to Eq. 3 (see Appendix C). This component can be computed from τ_w by Eq. 4 (see Appendix C). Unfortunately, none of the references gives details on how the wall shear stress was computed on the hill. We consider the wall-parallel definition of τ_w as more consistent, in particular also for the assessment of

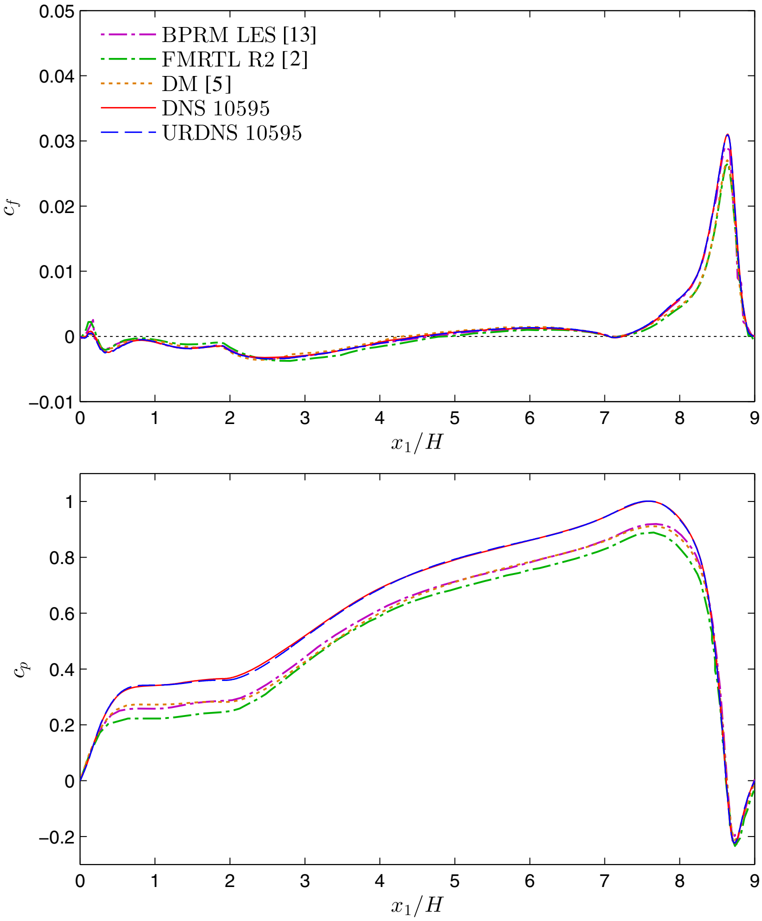

Fig. 7 $\mathrm{Re}_H = 10{,}595$: Skin friction (top) and pressure (bottom) coefficients at the lower wall

the near-wall resolution through the x_2^+ criterion, but $\tau_{w,1}$ may also be a possible choice if indicated clearly. Note that this issue is also present on the lee side of the hill, although not as clearly visible.

However, in the region between the hills, $2 \leq x_1/H \leq 7$, which is most interesting, the results are not affected by the difference in the prostprocessing as the wall is horizontal, allowing a comparison of the curves. The data sets URDNS 10595, DNS 10595, DM and BPRM LES show good accordance in the region from $x_1/H = 2$ to $x_1/H = 4$, but the data of DM exhibits minor oscillations, which may be due to the comparably short sampling time used in that work or due to a remaining discretization error (see Section 5). The case FMRTL R2 underpredicts the wall shear stress in that region. Near the reattachment zone in the interval $x_1/H = 4$ to $x_1/H = 6$, the accordance of DM and the present simulation cases is still good, whereas both FMRTL R2 and BPRM LES yield an underprediction. Accordingly, the primary recirculation bubble separates at around $x_{1,\mathrm{sep}}/H = 0.20$ and reattaches near $x_{1,\mathrm{reatt}}/H = 4.5$, where the two cases FMRTL R2 and BPRM LES predict larger values and the case DM a shorter reattachment length as compared to the other three cases. The measurement data specifies the reattachment length significantly shorter as $x_1/H = 4.21$. All available data is compared in Table 2. The secondary recirculation at the windward side of the hill separates at $x_1/H = 7.05$ and reattaches at $x_1/H = 7.24$ according to the case DNS 10595, which is in agreement with the reference data [13]. In addition to these two recirculations, a third recirculation may be observed at the hill crest according to the references [2, 13]. The separation point is computed to $x_1/H = 0.00$ and the reattachment to $x_1/H = 0.08$ based on the case DNS 10595. This recirculation is so thin that it is not visible in the mean velocity profiles presented in the subsequent section, where the first data point at $x_1/H = 0.05$ is located $0.0015/H$ above the wall. This may explain why this recirculation bubble was not found in the experiments in [15] at a similar Reynolds number. The excellent agreement between the cases DNS 10595 and URDNS 10595 in the whole range indicates that the spatial discretization error, as well as other uncertainties, indeed are negligible.

Next, we discuss the pressure coefficient also included in Fig. 7. The effect of the different length of the pressure ramp near $x_1/H = 0$ discussed in conjunction with the lower Reynolds number is even more severe in the present case. Again, this disagreement shifts the remaining part of the plot in vertical direction, as the reference pressure is considered at the lower wall at $x_1/H = 0$. The pressure coefficient profiles DNS 10595 and URDNS 10595 essentially lie on top of each other, whereas the cases BPRM LES, FMRTL R2 and DM exhibit a noticeable offset. The cases DNS 10595 and URDNS 10595 show an outstanding agreement throughout. The other simulation cases follow the trend well, despite the offset.

4.2.2 $Re_H = 10{,}595$: *velocity statistics*

Profiles of the averaged velocity, Reynolds shear stress, and turbulence kinetic energy are plotted in Fig. 8 for the Reynolds number $\mathrm{Re}_H = 10{,}595$ at nine stations. The present simulations are compared to the two LES reference cases, BPRM LES and FMRTL R2, as well as experimental results, RM Exp, where available. Regarding the mean streamwise velocity, the curves of the cases DNS 10595 and URDNS 10595 lie mostly in between the experimental (RM Exp) and LES references (BPRM LES and FMRTL R2), for example above the hill crest or in the reattachment zone, but the differences are very small overall. With respect to the vertical velocity, there are more noticeable aspects. On the lee side of the hill, there is a distinct gap between the LES cases BPRM LES and FMRTL R2 on the one hand and the present simulation results of DNS 10595 and URDNS 10595 on the other hand. The last station at $x_1/H = 8$ shows a similar picture, as the present simulation results lie in the gap between the LES and experimental references.

Regarding the profiles of the Reynolds shear stress, the cases BPRM LES and FMRTL R2 underpredict the magnitude of the stress in the thin shear layer in the range $0 \leq x_1/H \leq 1$ in comparison to the cases DNS 10595 and URDNS 10595. The experimental data is generally in better accordance with the numerical data than for the previously discussed Reynolds number of $\mathrm{Re}_H = 5{,}600$.

The profiles of the turbulence kinetic energy of the cases DNS 10595 and URDNS 10595 are in excellent agreement, which allows the conclusion that the spatial discretization of the DNS is capable of resolving all relevant scales. In the sharp shear layer above the recirculation bubble, both LES reference cases BPRM LES and FMRTL R2 predict considerably lower magnitudes. A possible explanation for these differences may be that the turbulent motions in this area are not sufficiently resolved. We note that this underresolution also may

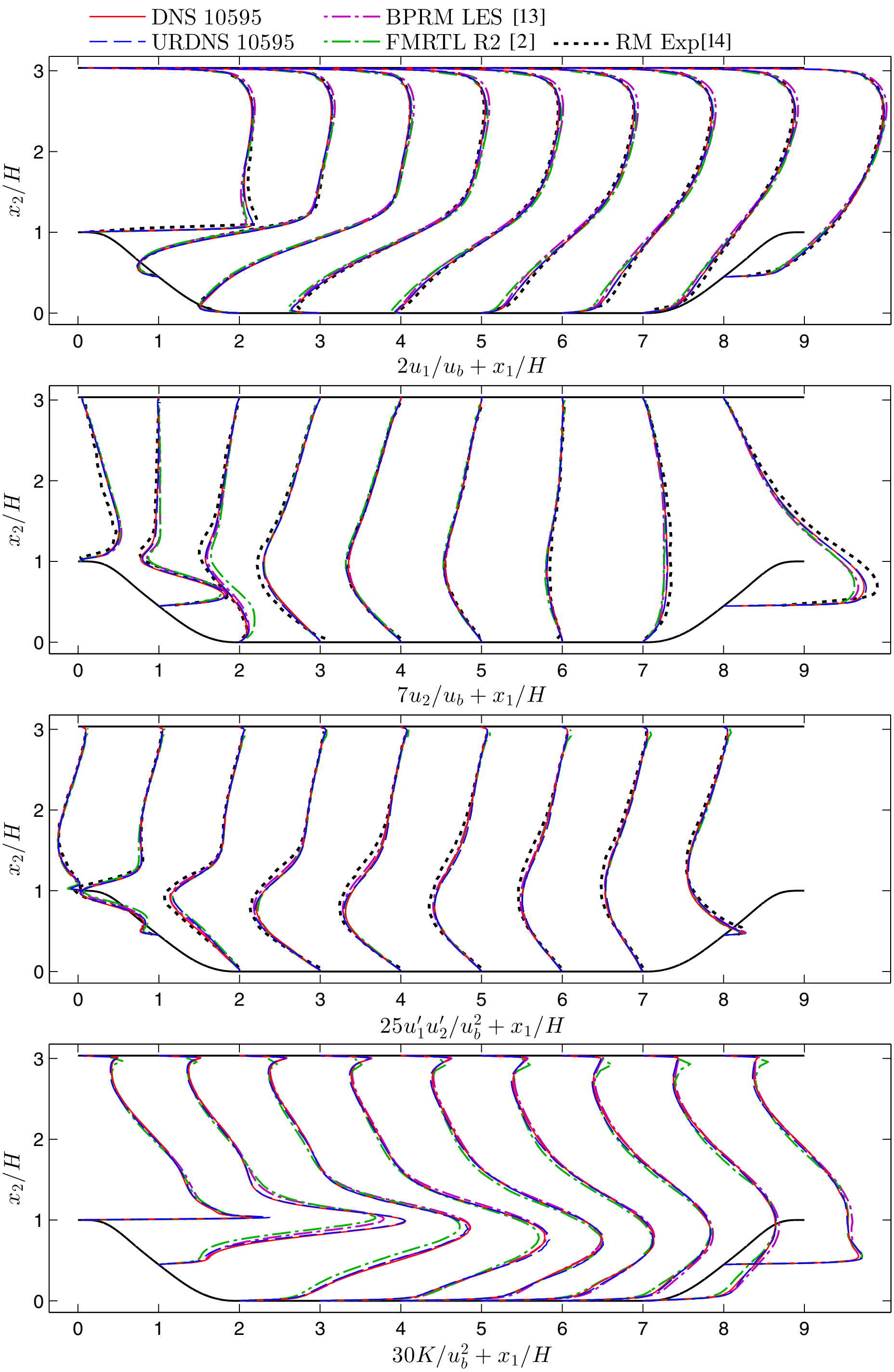

Fig. 8 $\mathrm{Re}_H = 10{,}595$: Profiles of mean velocity in streamwise and vertical direction u_1 and u_2, Reynolds shear stress $u_1'u_2'$, as well as turbulence kinetic energy $K = 1/2(u_1'u_1' + u_2'u_2' + u_3'u_3')$ (from top to bottom)

be the cause for the overestimation of the reattachment lengths by the two LES reference cases in comparison to the present DNS 10595.

The following major conclusions may be drawn from the discussion of the Reynolds number $\mathrm{Re}_H = 10{,}595$:

- the DNS 10595 and URDNS 10595 cases show excellent agreement in all figures, which indicates that the spatial discretization error as well as other sources of error are minor,
- the widely used reference data BPRM LES and FMRTL R2 may not capture all relevant turbulent motions in the thin shear layer above the recirculation bubble sufficiently well, which may come along with an underestimation of the momentum exchange in that region, resulting in a longer recirculation,
- the case DNS 10595 yields a shorter reattachment length than in the previously available LES reference data BPRM LES and FMRTL R2, but longer than in the experiments RM Exp.

4.3 Analysis of the averaging error

The averaging times considered in the reported simulation cases differ, according to Tables 1 and 2, by approximately one order of magnitude, which we see as an indication that this parameter should be investigated in more detail. Beginning with the mean velocity components, a frequently used measure for the averaging error in these quantities is the deviation of the spanwise velocity, which should vanish in the mean. The absolute values of the spanwise velocity profiles are consistently below 1%, most of the time even significantly below 0.5%; an error that seems to be acceptable.

However, it was already reported by Rapp [40] in the context of the measurements that it was difficult to determine the reattachment length of the primary recirculation bubble due to low-frequency motions of the reattachment point in streamwise direction. The recent experimental study [15] further elaborates on the issue of rare events having an influence on the flow statistics. This behavior motivates a separate analysis of the averaging error of the reattachment length. This length is of paramount importance in this benchmark case, as it is frequently used as an 'error norm', summarizing the overall performance of a numerical method or model in one number.

In order to analyze the statistical convergence behavior of this quantity in detail, we plot intermediate convergence results of $x_{1,\mathrm{reatt}}/H$ over the respective sampling time in the number of flow-through times $\Delta t_{\mathrm{aver}}/T^*$, see Fig. 9 for $\mathrm{Re}_H = 5{,}600$ and Fig. 10 for $\mathrm{Re}_H = 10{,}595$. These data points are subject to an error bandwidth, which converges as $\sim \pm\mathrm{const}/\sqrt{\Delta t_{\mathrm{aver}}/T^*}$ according to the statistical theory of the 'standard error of the mean'. In order to estimate the error at the end of the sampling time, this error bandwidth is also plotted in Figs. 9 and 10 by considering the curves $x_{1,\mathrm{reatt}}/H \pm \alpha/\sqrt{\Delta t_{\mathrm{aver}}/T^*}$ and adjusting the constant parameter α manually such that most points lie within the bandwidth. This procedure yields an estimation of the sampling error for $x_{1,\mathrm{reatt}}/H$ of ± 0.09 for $\mathrm{Re}_H = 5{,}600$ and ± 0.06 for $\mathrm{Re}_H = 10{,}595$.

Both plots include the available reference data, which may be compared to the error bandwidth of the DNS cases. Regarding $\mathrm{Re}_H = 5{,}600$, the references BPRM DNS/LES lie within the bandwidth, which means that they are in agreement with the present DNS within the available accuracy. The experimental reattachment length and the case URDNS 5600 lie outside of the bandwidth, rendering other sources of error than the statistical averaging error

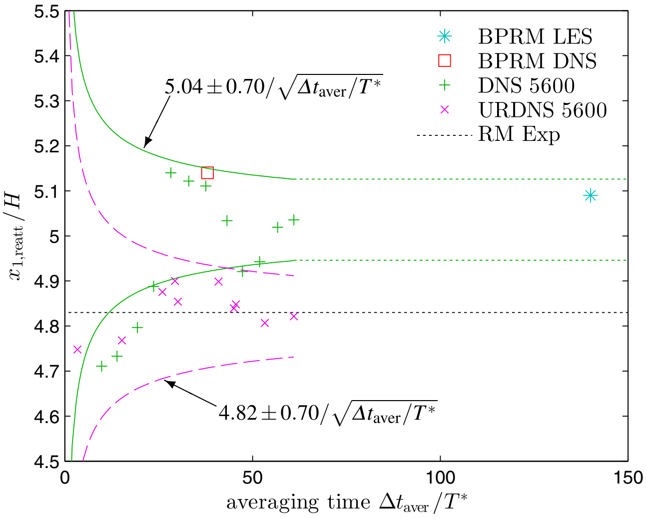

Fig. 9 $\mathrm{Re}_H = 5{,}600$: Analysis of the statistical averaging error of the reattachment length $x_{1,\mathrm{reatt}}/H$. The error bandwidth is given as $x_{1,\mathrm{reatt}}/H \pm \mathrm{const}/\sqrt{\Delta t_{\mathrm{aver}}/T^*}$ and yields an estimation of the error after 61 flow-through times of ± 0.09

relevant. Regarding $\mathrm{Re}_H = 10{,}595$, it is apparent that the bandwidths of URDNS 10595 and DNS 10595 overlap significantly, such that the remaining difference in the reattachment length may be a statistical averaging error. This result mirrors the excellent agreement of the data discussed in the previous section. The cases BPRM LES and FMRTL R2 lie well outside of the error bandwidth, indicating that the differences to the DNS are other sources of error, such as underresolution. The experimental reference also lies significantly outside

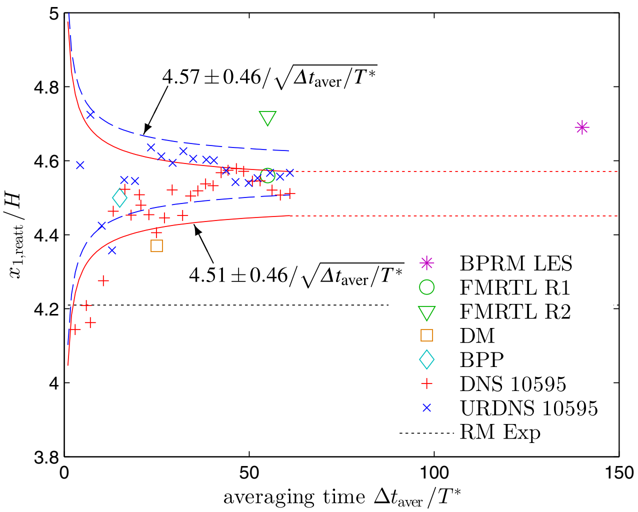

Fig. 10 $\mathrm{Re}_H = 10{,}595$: Analysis of the statistical averaging error of the reattachment length $x_{1,\mathrm{reatt}}/H$. The error bandwidth is given as $x_{1,\mathrm{reatt}}/H \pm \mathrm{const}/\sqrt{\Delta t_{\mathrm{aver}}/T^*}$ and yields an estimation of the error after 61 flow-through times of ± 0.06

of the uncertainty range. The cases BPP, FMRTL R1, and DM lie inside or near the bandwidth, such that these cases are in agreement with the DNS within the available accuracy, which does not exclude other sources of error, however.

It may be concluded from this discussion that the reattachment length of this flow is very sensitive to insufficient sampling. This fact may be explained by the small slope of the c_f curves in Figs. 5 and 7 near the reattachment points. While the statistical error after 61 flow-through times of $\pm 0.09H$ and $\pm 0.06H$ may seem large as reference data, the reduction by a factor of two for the higher Reynolds number would require an additional sampling time of 183 flow-through times due to the slow convergence with $\sim 1/\sqrt{\Delta t_{\text{aver}}/T^*}$, which would require additional 19 million CPU core hours and three months of computation time, as well as 400,000 kWh of electricity. Furthermore, sampling times as high as 1000 flow-through times suggested in [15] will not be available in the context of DNS at the present Reynolds numbers for several years, nor are such sampling times desirable in the application of benchmark simulations. As a result, researchers have to be made aware of the presence of statistical errors and they should be taken into account when using the DNS data as a reference.

5 Detailed h/p Convergence Analysis and Assessment of Implicit Large-Eddy Simulation

The high computational cost of the DNS makes an application of this approach to most industrial flows infeasible. Computing only the larger, inhomogeneous eddies and modeling the smaller, homogeneous turbulent motions promises a similar level of accuracy for the present benchmark example at a drastically reduced numerical effort. In this section, we analyze the speed of convergence towards the DNS at $\mathrm{Re}_H = 10{,}595$ in a detailed h/p convergence study. To this end, the coarsening of the mesh, starting from the DNS, gradually yields URDNS and ILES.

The scientific community is divided about the question if ILES is a viable approach for high-order DG schemes or if explicit subgrid models should be used. For many years, researchers have argued that ILES in conjunction with the high-order DG method is highly efficient, see, e.g., [4, 27]. Recently, several studies concluded that the subgrid dissipation of high-order DG methods is not sufficient at high Reynolds numbers, see, e.g., [41, 42] or that only relatively low polynomial degrees (k = 2) provide sufficient dissipation at very high Reynolds numbers to achieve stable results at the price of a lower level of accuracy [43]. While the numerical dissipation at very high polynomial degrees ($k > 10$) indeed may be inappropriate for ILES, very good results using the present scheme have been obtained with moderately high polynomial degrees $3 \leq k \leq 5$ [6, 8, 16, 34]. For example, the present scheme has been used for wall-modeled ILES of turbulent channel flow up to $\mathrm{Re}_\tau = 5{,}200$ and flow over periodic hills up to $\mathrm{Re}_H = 37{,}000$ and excellent results have been obtained using $k = 4$ [8]. In addition, the study in [44] presents very promising results of wall-modeled ILES of turbulent channel flow up to $\mathrm{Re}_\tau = 50{,}000$ using a high-order DG method with $k = 4$ as well. Due to these arguments for ILES, we do not consider an explicit subgrid model in the underresolved simulations in the present convergence study.

Finally, we note that the present scheme is very robust and does in general not exhibit any stability or aliasing problems. The main reason for the excellent stability behavior is the use of exact overintegration in combination with the formulation using consistent penalty terms. From a computational point of view, the dual splitting scheme applied to the *incompressible* Navier–Stokes equations allows to apply overintegration to the nonlinear convective step only, and thus essentially for free, see also the analysis in [34]. Conversely, a *compressible* solver must in general advance all terms with overintegration. This issue is exacerbated by a higher degree of nonlinearity at moderate "artificial" Mach numbers which are often chosen at around 0.1, see, e.g., [21]. As a consequence, stability is often sacrificed for a better performance in the *compressible* case by using an insufficient number of integration points.

Regarding the h convergence study, we consider meshes of 32×16×16 (coarse mesh), 64×32×32 (medium mesh), and 128×64×64 (fine mesh) cells, all with $k = 4$, since this polynomial degree appears to provide an appropriate amount of subgrid dissipation; see the aforementioned references. With respect to p convergence, we take the medium mesh, 64×32×32, and vary the polynomial degree through $k = 3$, 4, 5, and 6. In order to achieve a full h/p analysis, we present further results of the coarse and fine mesh with all four polynomial degrees as well as two additional results for $k = 2$, which are only evaluated by comparing their reattachment length as a global measure of the error. The total number of grid points for these meshes is obtained by taking the number of cells times $k + 1$ in each dimension. All simulation cases are listed in Table 3. Note that the two finest cases are equal to the DNS 10595 and URDNS 10595 cases presented in the previous sections. The grid stretching is adjusted in order to achieve a compromise between near-wall and bulk resolution. The first off-wall Gauss–Lobatto grid point is plotted in Fig. 11. Representative

Table 3 Detailed h/p convergence study at $\mathrm{Re}_H = 10{,}595$: Simulation cases and reference data, including polynomial degree k, order of accuracy $(k + 1)$, total number of grid points N, Mach number Ma/incompressible (inc.), maximum value of Δx_2^+ at the upper wall (near-wall resolution) and h/η (bulk flow resolution), sampling time Δt_{aver} in number of flow-through times T^*, separation and reattachment length $x_{1,\mathrm{sep}}$ and $x_{1,\mathrm{reatt}}$

Ref.	Grid points	k	Order	N	Ma	$\max\left(\Delta x_2^+\right)$	$\max\left(\frac{h}{\eta}\right)$	$\frac{\Delta t_{\mathrm{aver}}}{T^*}$	$\frac{x_{1,\mathrm{sep}}}{H}$	$\frac{x_{1,\mathrm{reatt}}}{H}$
–	128×64×64	3	4	0.52M	inc.	5.69	17.3	61	0.27	3.65
–	160×80×80	4	5	1.0M	inc.	3.28	15.4	61	0.23	3.90
–	192×96×96	5	6	1.8M	inc.	2.21	13.8	61	0.22	4.19
–	224×112×112	6	7	2.8M	inc.	1.60	12.8	61	0.21	4.19
–	192×96×96	2	3	1.8M	inc.	4.89	12.4	61	0.25	3.58
–	256×128×128	3	4	4.2M	inc.	2.62	11.7	61	0.21	3.81
–	320×160×160	4	5	8.2M	inc.	1.66	9.64	61	0.21	4.22
–	384×192×192	5	6	14.2M	inc.	1.30	8.48	61	0.21	4.24
–	448×224×224	6	7	22.5M	inc.	1.09	7.55	61	0.21	4.38
–	384×192×192	2	3	14.2M	inc.	3.19	6.92	61	0.19	4.01
–	512×256×256	3	4	33.6M	inc.	1.75	6.17	61	0.20	4.23
–	640×320×320	4	5	65.5M	inc.	1.10	5.36	61	0.19	4.36
–	768×384×384	5	6	113M	inc.	0.86	4.65	61	0.19	4.57
–	896×448×448	6	7	180M	inc.	0.66	4.07	61	0.20	4.51
[4]	160×80×80[a]	4	–	1.0M	0.1	–	–	56	–	3.62[b]
[4]	224×112×112[a]	6	–	2.8M	0.1	–	–	56	–	4.18[b]
[7]	256×128×128	3	4	4.2M	0.1	–	–	–	–	3.9

The statistical averaging uncertainty of $x_{1,\mathrm{reatt}}$ is approximately ±0.06 in all cases. The number of nodes of the present DG solver is obtained by the number of cells in each spatial direction times the number of nodes per cell $(k + 1)$ in each direction

[a]Private communication with Thomas Bolemann

[b]Extracted from figure 9 in [4]

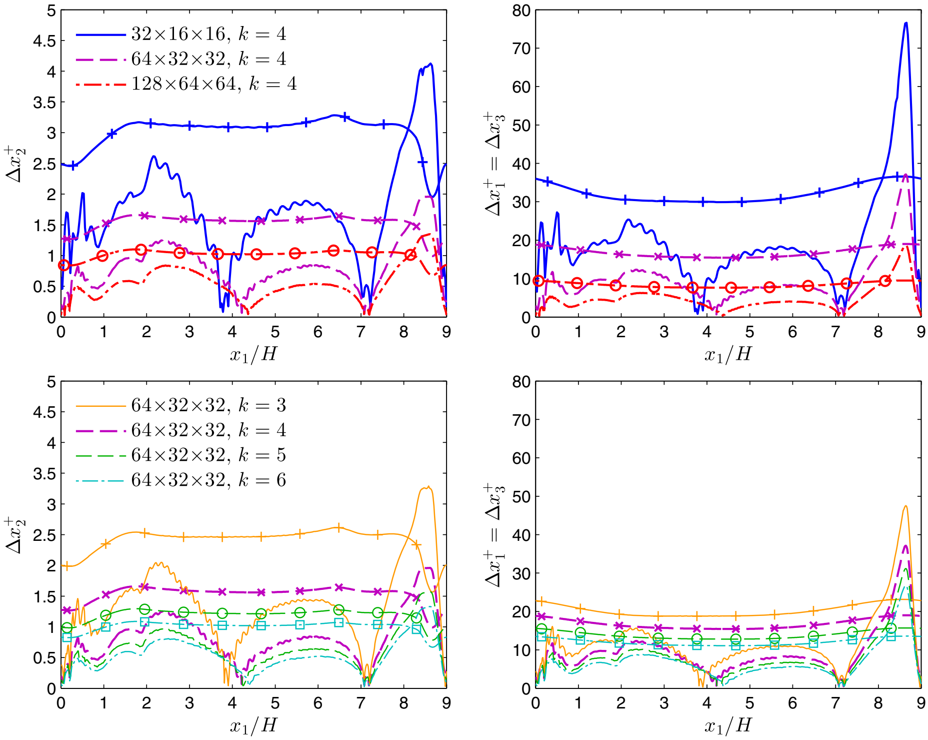

Fig. 11 ILES h/p convergence study at $\mathrm{Re}_H = 10,595$: Location of first off-wall Gauss–Lobatto point Δx_2^+ (left) as well as normalized grid spacings in streamwise and spanwise direction $\Delta x_1^+ = \Delta x_3^+$ (right). The curves corresponding to the upper wall include markers

mesh parameters are also included in Table 3, given as the maximum value of Δx_2^+ at the upper wall, and the maximum ratio of h/η occurring in the shear layer similar to Fig. 4. The results are compared to the case DNS 10595 discussed in Section 4.2. We additionally compare the results of the reattachment length to previous ILES by [4] and LES (based on the WALE model) by [7] using *compressible* high-order DG methods with comparable spatial discretizations, which are also included in Table 3.

5.1 ILES h/p convergence study at $\mathrm{Re}_H = 10{,}595$: skin friction and pressure coefficients

We present the skin friction and pressure coefficients for the h convergence study in Fig. 12 and for the p convergence study in Fig. 13. Regarding the h convergence, the skin friction on the windward side of the hill exhibits differences to the DNS for the coarse mesh only and is converged for the medium and fine mesh. On the lee side of the hill, the coarse mesh yields significant deviations from the reference, whereas the medium and fine meshes agree well with the DNS. In the recirculation zone between $x_1/H = 3$ and $x_1/H = 5$, the skin friction coefficient is overpredicted by the coarse and medium simulations. In addition, the coarse case exhibits characteristic waves, which seem to arise due to the construction of the present high-order DG discretization and have been observed in previous ILES studies using *compressible* high-order DG methods [4, 5] as well. The waves may be observed with

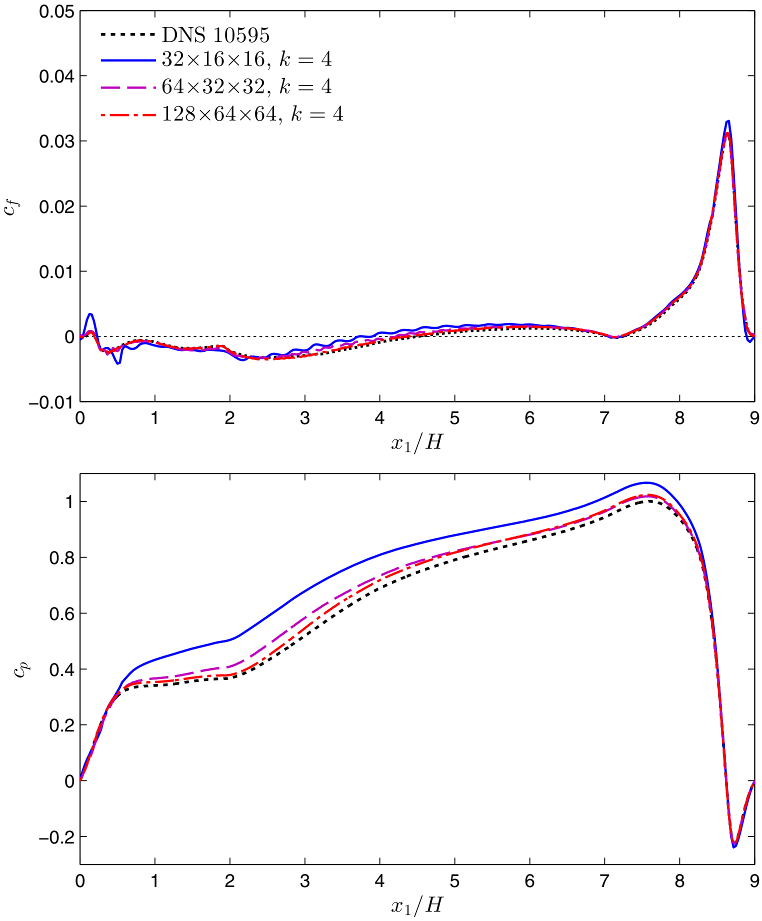

Fig. 12 ILES h convergence study at $\mathrm{Re}_H = 10{,}595$: Skin friction (top) and pressure (bottom) coefficients at the lower wall

reduced amplitude for the medium mesh and are not visible for the fine mesh. It was found in [4] that this effect is not related to insufficient statistical averaging. It is interesting to see that [7] presents smooth curves in the simulations using the WALE subgrid model, and it should be further investigated if this improvement is indeed achieved by the use of an eddy viscosity subgrid model. With respect to the pressure coefficient, the curves converge quickly, with significant deviations for the coarse case, moderate differences for the medium case and minor errors for the fine case.

The p refinement shows generally a similar trend as the h refinement, and exhibits a smaller bandwidth regarding the error level. The skin friction of the case $k = 5$ shows minor errors and the case $k = 6$ is almost indistinguishable from the DNS. The waviness of the skin friction near the recirculation zone is also present with a smaller amplitude, but vanishes incrementally for the higher degrees. The pressure curves improve significantly from $k = 3$ to $k = 4$, but the finest case still differs from the DNS.

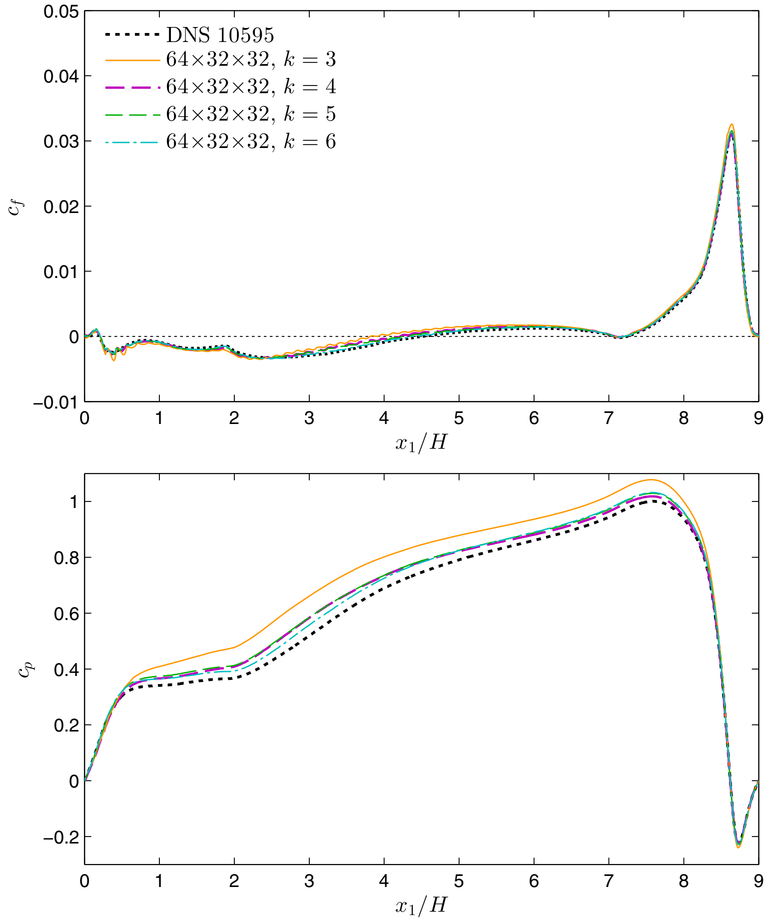

Fig. 13 ILES p convergence study at $\mathrm{Re}_H = 10{,}595$: Skin friction (top) and pressure (bottom) coefficients at the lower wall

The reattachment lengths of all cases of the h/p convergence analysis are given in Table 4. It is noted that the reattachment length is consistently underpredicted by the coarse meshes, in accordance with the overprediction of the skin friction coefficient in that region, and most cases with a reasonably fine LES grid lie in the area around $x_1/H = 4.2$. While this result is quite good, it also indicates that a further improvement of the ILES approach is possible. The ILES and LES references [4, 7] listed in Table 3, which use *compressible* high-order DG methods, show almost the same reattachment lengths as the simulation case with the corresponding mesh and polynomial degree.

The question of which polynomial degree is most efficient for underresolved simulations with the present scheme is difficult to answer in general. Higher orders seem to perform better per degree of freedom in coarse simulation cases, as may be observed in Table 3. For example, comparing the cases of $k = 2$ with the corresponding cases of $k = 5$ and one h refinement level coarser (with the same number of degrees of freedom), it may be observed that the cases of $k = 5$ predict the reattachment length much better. As the simulation time

Table 4 ILES h/p convergence study at $\mathrm{Re}_H = 10{,}595$: Summary of reattachment lengths $x_{1,\mathrm{reatt}}/H$

$N_{e1}\times N_{e2}\times N_{e3} \setminus k$	2	3	4	5	6
32×16×16		3.65	3.90	4.19	4.19
64×32×32	3.58	3.81	4.22	4.24	4.38
128×64×64	4.01	4.23	4.36	4.57	4.51

Data is redundant with Table 3, and reprinted using number of cells N_{ei} per spatial direction i and polynomial degree k. The statistical averaging uncertainty is approximately ± 0.06 in all cases

rather than number of grid points is a more relevant measure for the computational cost, the number of required core hours has been evaluated for all simulation cases and is displayed in Table 5; see Appendix B for details on the computational setup. It may be observed that the computational cost of the case $k = 2$ using the medium mesh is by a factor of 3 lower compared to the case $k = 5$ using the coarse mesh. Analogously, the computational cost of the case $k = 2$ (fine mesh) is by a factor of 2.5 computationally cheaper than the case $k = 5$ (medium mesh). The higher level of accuracy of the $k = 5$ cases overcompensates the higher computational cost. Comparing the other results presented in Tables 4 and 5, the polynomial degree of $k = 4$ generally seems to provide a good compromise between accuracy and computational cost.

5.2 ILES h/p convergence study at $\mathrm{Re}_H = 10{,}595$: velocity statistics

The cases considered in the h/p convergence study are further compared via profiles at nine streamwise stations in Figs. 14 and 15. Regarding the streamwise velocity, the agreement with the DNS is generally very good. Solely near the lower wall in the region of the reattachment point, the velocity is overpredicted by the coarser simulation cases in accordance with the shorter recirculation bubble observed in these cases. Regarding the vertical velocity, the coarser cases show differences to the DNS, but the simulations show a clear trend to convergence with increasing resolution. With respect to the Reynolds shear stresses, the errors are on a very low level throughout and deviations from the DNS are only significant for the coarsest case in the h and p convergence study, respectively. The turbulence kinetic energy profiles agree well with the DNS, again except for the coarsest cases, which show some deviations. In addition, the profiles of the coarsest simulations exhibit unphysical peaks at the element boundaries, which originate from the discontinuity present in the solution at the cell boundaries and have been reported in earlier studies as well (see, e.g., [7]). The study in [7] found that this issue is considerably improved if an eddy viscosity subgrid model is

Table 5 ILES h/p convergence study at $\mathrm{Re}_H = 10{,}595$: Computational cost in thousands of core hours, i.e., the number of MPI ranks times wall time

$N_{e1}\times N_{e2}\times N_{e3} \setminus k$	2	3	4	5	6
32×16×16		0.94	3.1	13	43
64×32×32	4.3	18	74	190	700*
128×64×64	77	310*	920*	2,700*	8,700*

The cases marked with * were performed on Phase 1 of SuperMUC, consisting of Intel Sandy Bridge CPUs at 2.7GHz, and the remaining simulations on a cluster using Intel Haswell CUPs at 2.5 GHz (see Appendix B)

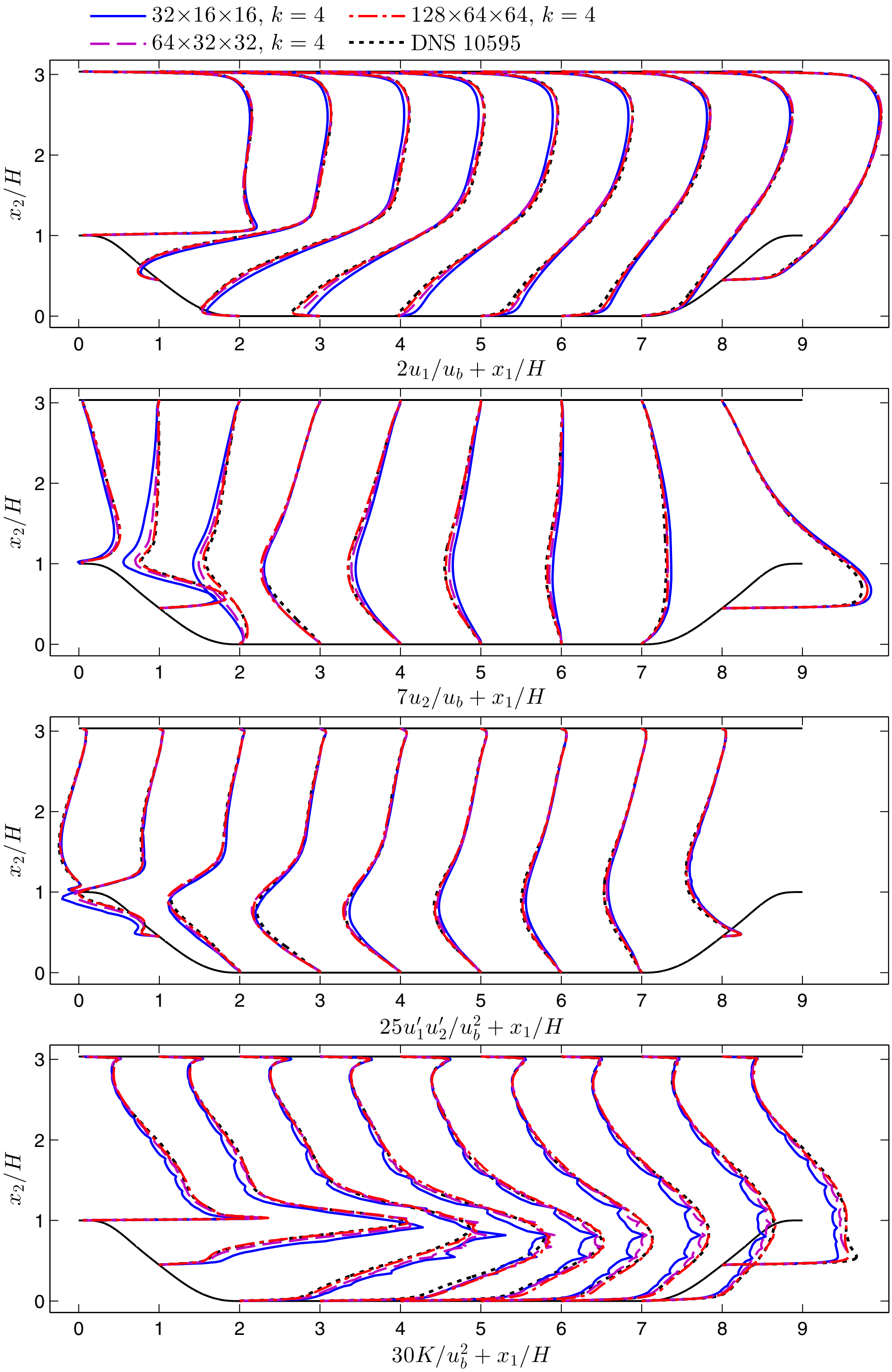

Fig. 14 ILES h convergence study at $\mathrm{Re}_H = 10{,}595$: Profiles of mean velocity in streamwise and vertical direction u_1 and u_2, Reynolds shear stress $u_1'u_2'$, as well as turbulence kinetic energy K (from top to bottom)

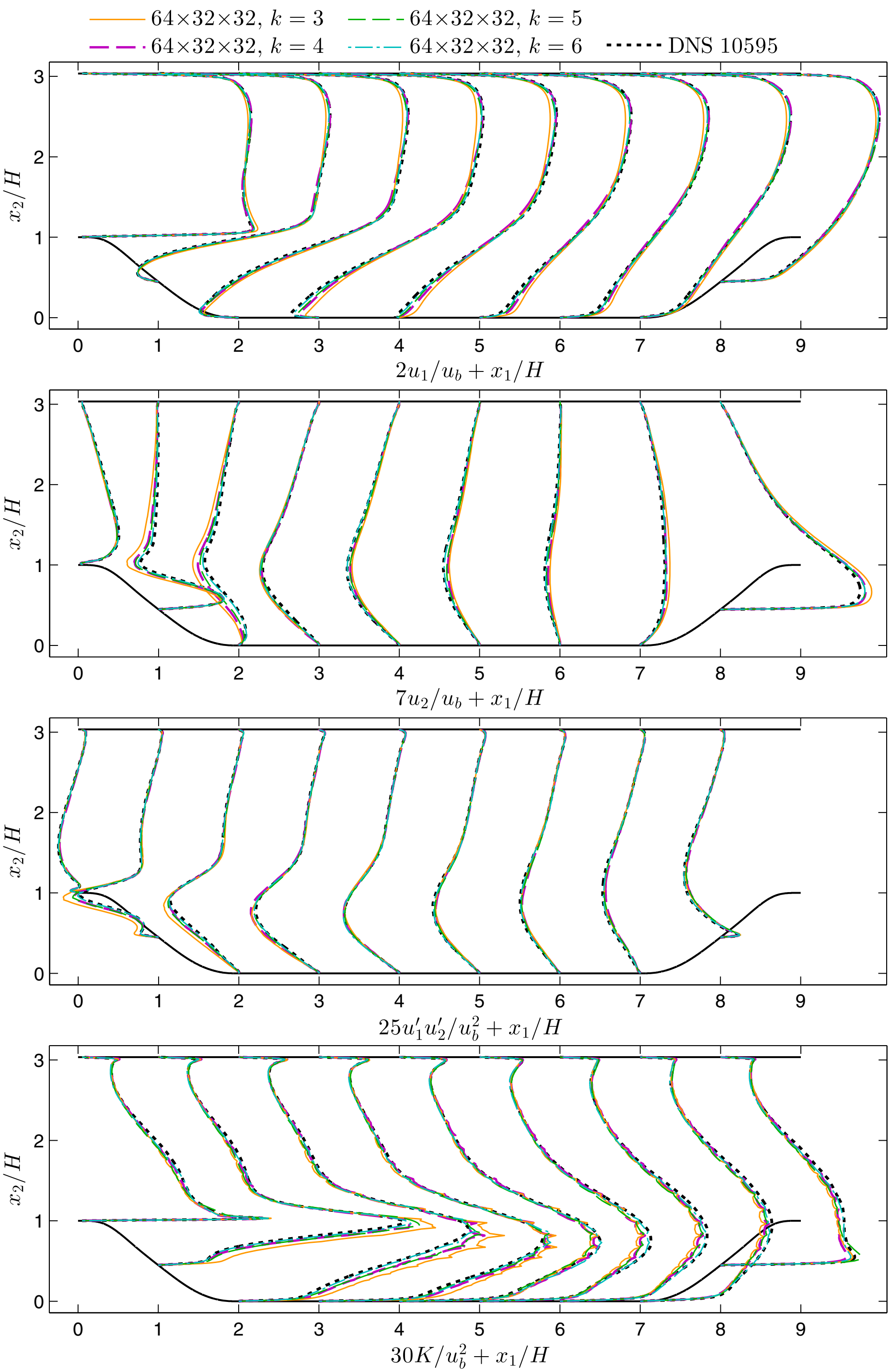

Fig. 15 ILES p convergence study at $\mathrm{Re}_H = 10{,}595$: Profiles of mean velocity in streamwise and vertical direction u_1 and u_2, Reynolds shear stress $u_1'u_2'$, as well as turbulence kinetic energy K (from top to bottom)

included. The present results confirm the observations made in [7] that these peaks vanish with increasing resolution and the DNS is completely smooth.

The present convergence analysis has shown the speed of convergence of the present setup towards the DNS results in a detailed h/p refinement study. The results show that the methodology yields a high level of accuracy in underresolved turbulence simulations, although the length of the recirculation region is predicted too short in the coarsest simulations. Moderate polynomial degrees near $k = 4$ yield a good compromise between accuracy and time to solution and the numerical fluxes seem to provide an appropriate amount of subgrid dissipation at this degree.

6 Concluding Remarks

In this work, we have presented DNS of flow over periodic hills at the hill-Reynolds numbers $\mathrm{Re}_H = 5{,}600$ and $\mathrm{Re}_H = 10{,}595$. The simulations were performed using spectral *incompressible* discontinuous Galerkin schemes of 8^{th} and 7^{th} order of spatial accuracy, respectively. The DNS at the lower Reynolds number is in very good agreement with a DNS reference, but the new results do not show oscillations in the curves of the skin friction coefficient. The DNS at the higher Reynolds number is considered to be of much higher fidelity than the previous numerical reference data, as we have used more than twice the number of grid points in each spatial direction, in addition to the high-order accuracy. The results of the higher Reynolds number lie in between the experimental and the LES reference data available in the literature. A detailed h/p convergence analysis at the higher Reynolds number has shown that the errors associated with the DNS are small.

The DNS results have been made available on a public open source repository [38] and may be used as reference data. The data sets include the averaged skin friction and pressure coefficients c_f and c_p on the lower and upper wall, as well as profiles of the mean velocity in streamwise and vertical direction u_1 and u_2, the Reynolds stresses $u_1'u_1'$, $u_2'u_2'$, $u_3'u_3'$, $u_1'u_2'$ (and turbulence kinetic energy $1/2(u_1'u_1' + u_2'u_2' + u_3'u_3')$). Regarding the skin friction profiles, we detailed in this work how the value was computed from the simulated velocity field in order to allow an accurate comparison. With respect to the pressure coefficient curves, we argue that a value for the reference pressure at the upper wall would be better suited than the widely used reference pressure on the hill crest. When comparing the reattachment length of this flow, the statistical averaging error has to be considered, as it is relevant even for very long averaging times. The numerical results of the present work have yielded reattachment lengths of $x_1/H = 5.04 \pm 0.09$ for $\mathrm{Re}_H = 5{,}600$ and $x_1/H = 4.51 \pm 0.06$ for $\mathrm{Re}_H = 10{,}595$.

Finally, we have shown that the present numerical method may be employed for ILES as well. We performed a detailed h/p convergence study at $\mathrm{Re}_H = 10{,}595$ and yielded good agreement with the DNS in a wide range of spatial resolution.

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Compliance with Ethical Standards

Conflict of interests The authors declare that they have no conflict of interest.

Appendix A: Constant Mass Flux Control

The flow is driven by a body force $\boldsymbol{f} = (f_1, 0, 0)^T$. In this work, we use a simple proportional control algorithm with an additional damping term to ensure a constant mass flux. We define

$$f_1^{n+1} = f_1^n + \beta_1(Q^0 - Q^n) - \beta_2(Q^n - Q^{n-1}), \tag{1}$$

where Q is the volume flux, β_1 the gain of the proportional controller and β_2 of the damper. If we would choose $\beta_1 = \beta_2 = 1/\Delta t$, the controller would be identical to the frequently used algorithm in [35]. In practice, we have found that another empirical choice of β_1 and β_2 increases the robustness of the approach. For the present computations, we choose $\beta_1 = 500$ and $\beta_2 = 30{,}000$, corresponding to $H = 0.028$, $\rho = 1$, and $u_b = 5.621$, yielding $Q^0 = 0.040376$. With this choice, the mass flux is constant to five digits of accuracy after an initial transient.

Appendix B: Computational Cost

The DNS were performed on Phase 1 of SuperMUC in Garching, Germany, consisting of 2×8 core Intel Sandy Bridge CPUs at 2.7GHz. Scaling experiments for both cases are shown in Fig. 16. At $\mathrm{Re}_H = 5{,}600$, the solver exhibits very good strong scaling up to 2,048 CPU cores and the curve flattens up to 16,384 CPU cores. The case $\mathrm{Re}_H = 10{,}595$ yields excellent strong scaling up to 8,192 CPU cores and shows a speed-up until 65,536 CPU cores. All steps of the time integration scheme show close-to-optimal scaling, except for the pressure Poisson algorithm, which involves a significant amount of communication [24]; the scaling of the pressure Poisson solver is also included in Fig. 16. Regarding the simulation at $\mathrm{Re}_H = 10{,}595$, the shortest absolute wall time achieved in this scaling test is approximately 0.22s per time step at 65,536 compute cores. As a good compromise between resource efficiency and run time, the final computations were performed on 8,192 cores, yielding a wall time per time step T_{wall} of approximately 0.45s and an overall run time of approximately six weeks for 8.5 million time steps. All simulations that required a computational cost of less than approximately 200 thousand core hours were performed on an in-house cluster, consisting of 2×12 core Intel Haswell CPUs E5-2680v3 at 2.5GHz.

Appendix C: Wall Shear Stress on Curved Lower Boundary

At the lower wall, where the boundary is curved, the wall-parallel shear stress τ_w is computed from the Cartesian velocity components $\boldsymbol{u} = (u_1, u_2, u_3)^T$ via the velocity gradient gradient as

$$\tau_w = \operatorname{sgn}\left(\frac{\partial u_1}{\partial x_2}\right)\rho\nu\sqrt{\left(\frac{\partial u_1}{\partial x_1}n_1 + \frac{\partial u_1}{\partial x_2}n_2\right)^2 + \left(\frac{\partial u_2}{\partial x_1}n_1 + \frac{\partial u_2}{\partial x_2}n_2\right)^2} \tag{2}$$

and using the components of a wall-normal vector $\boldsymbol{n} = (n_1, n_2, n_3)^T$ of unit length, and the sign function sgn. Note that this formulation is equivalent to the alternative $\tau_w = \rho\nu\boldsymbol{t}^T \cdot (\nabla\boldsymbol{u} \cdot \boldsymbol{n})$ (evaluated in the x_1x_2-plane), with the tangential vector $\boldsymbol{t}$, in the continuous sense. Some studies seem to consider the streamwise velocity component only, yielding

$$\tau_{w,1} = \rho\nu\left(\frac{\partial u_1}{\partial x_1}n_1 + \frac{\partial u_1}{\partial x_2}n_2\right). \tag{3}$$

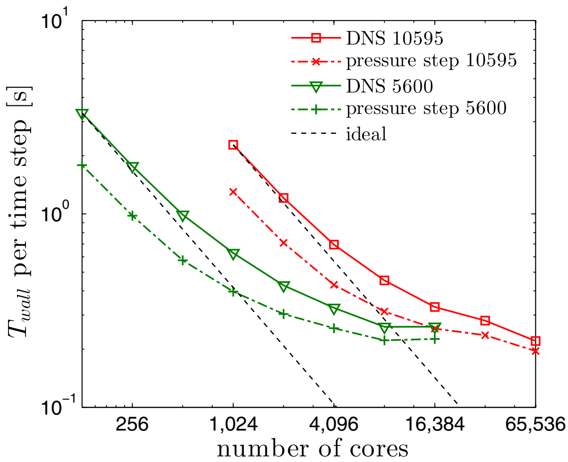

Fig. 16 Parallel scaling of the DNS setups at both Reynolds numbers up to 65,536 CPU cores

The definitions τ_w and $\tau_{w,1}$ are related by the slope $\frac{\mathrm{d}g}{\mathrm{d}x_1}$ of the function defining the lower wall, $g(x_1)$, by

$$\tau_{w,1} = \frac{\tau_w}{\sqrt{1 + \left(\frac{\mathrm{d}g}{\mathrm{d}x_1}\right)^2}}. \tag{4}$$

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